epi_comparison: BI next to a common information measure

What this folder shows

Perceived information (PI) is a common way to score how much a model knows about the key. It is sensitive to how confident the model is, not just to whether its guesses are right. A corrected version, effective PI (EPI), first rescales the model's confidence.

This folder compares BI with EPI on the same frozen model. Its main table (quoted in Section 5.2) rescales the model's confidence by factors from 0.05 to 20. Raw PI changes by tens of bits. EPI does not change at all, and no guess changes, so the count behind BI does not change either. Level D checks this table.

`epi_head_to_head.py` works on one trace set: it picks the most accurate of the 80 models, fits EPI's one parameter on separate traces, reports PI, EPI and BI with error bars, then repeats EPI at nine rescaling factors. `summarize_epi_head_to_head.py` combines the results of the six trace sets.

Before you start

The Docker image (or the Python environment with `requirements-ml.txt`) and the TCHES20 download.

Run

```
T=datasets/raw/tches20/TCHES20V3_CNN_SCA
python -m epi_comparison.epi_head_to_head --dataset ascad_desync_0 \
  --dataset-dir $T/datasets --tches20-src-dir $T/src --models-dir $T/models \
  --output-dir results/epi_comparison/main
```

```
$ T=datasets/raw/tches20/TCHES20V3_CNN_SCA; python -m epi_comparison.epi_head_to_head --dataset ascad_desync
    _0 --dataset-dir $T/datasets --tches20-src-dir $T/src --models-dir $T/models --output-dir results/epi_co
    mparison/main
[... earlier output omitted; key lines kept ...]
[...]
  "metric_scope": {
    "BI": "exact min-entropy leakage lower bound on uniform finite evaluation records",
    "EPI": "IUH+22 power-optimized Shannon cross-information; beta fit on disjoint calibration records"
  },
  "artifacts": {
    "model_selection": "model_selection.csv",
    "temperature_stress": "temperature_stress.csv",
    "calibration_scarcity": "calibration_scarcity.csv",
    "heldout_scores": "heldout_scores.npz"
  },
  "reproducibility": {
    "seed": 20260721,
    "beta_max": 64.0,
    "selection_fraction_per_class": 0.2,
    "calibration_fraction_per_class": 0.2,
    "bootstrap_reps": 2000,
    "scarcity_reps": 20,
    "hostname": "0c52c0f13b9f",
    "platform": "Linux-6.8.0-136-generic-x86_64-with-glibc2.36",
    "python": "3.11.9 (main, Sep  4 2024, 23:15:21) [GCC 12.2.0]",
```

```
    "numpy": "1.26.4",
    "tensorflow": "2.15.0",
    "slurm_job_id": "",
    "slurm_array_task_id": "",
    "elapsed_sec": 41.4214882850647
  }
}

100%|██████████| 10000/10000 [00:02<00:00, 4803.16it/s]
[exit code 0, 42 s]
```

```
$ cat results/epi_comparison/main/ascad_desync_0/temperature_stress.csv
temperature,raw_pi_bits,beta_hat,epi_bits,argmax_preserved,finite_bi_bits
0.05,-4.228038955965687,0.03890159485379491,0.006212571582823293,True,0.45663840374561676
0.1,-1.1297518927499726,0.07780318939675544,0.006212571593507192,True,0.45663840374561676
0.25,-0.1506471206644623,0.19450797411375473,0.006212571584957352,True,0.45663840374561676
0.5,-0.01945563916608881,0.38901593920225364,0.006212571647000034,True,0.45663840374561676
1.0,0.004733565821951712,0.7780318667398504,0.006212571687093431,True,0.45663840374561676
2.0,0.006038962604556953,1.5560637401463675,0.006212571675636203,True,0.45663840374561676
4.0,0.003944249255390852,3.1121274869594013,0.006212571669907591,True,0.45663840374561676
10.0,0.001800477850502584,7.780318545358356,0.006212571729040694,True,0.45663840374561676
20.0,0.0009374369681294083,15.560637096410725,0.006212571728062128,True,0.45663840374561676
[exit code 0, 0 s]
```

Run the other five trace sets the same way (`ascad_desync_50`, `ascad_desync_100`, `aes_hd`, `aes_rd`, `dpav4`). Level D leaves all six under `results/paper_results/recomputed/epi_comparison/main`, so you can also summarize those:

```
python -m epi_comparison.summarize_epi_head_to_head \
  --input-dir results/paper_results/recomputed/epi_comparison/main --output-dir results/epi_comparison/summary
cat results/epi_comparison/summary/summary.csv
```

```
$ python -m epi_comparison.summarize_epi_head_to_head --input-dir results/paper_results/recomputed/epi_compa
    rison/main --output-dir results/epi_comparison/summary && \
    cat results/epi_comparison/summary/summary.csv
# BI versus IUH+22 EPI at native long dimensions

Selection, beta calibration, and evaluation are class-balanced and mutually disjoint. BI and EPI use the sam
    e selected frozen model. BI is an exact finite-benchmark min-entropy lower bound; EPI is hel …

| Dataset | Native d | Eval n | Accuracy (Wilson 95%) | finite BI | raw PI | EPI | beta |
|---|---:|---:|---:|---:|---:|---:|---:|
| ascad_desync_0 | 700 | 24064 | 0.005361 [0.004514, 0.006366] | 0.4566 | 0.0047 | 0.0062 | 0.778 |
| ascad_desync_100 | 700 | 24064 | 0.014919 [0.013462, 0.016529] | 1.9333 | 0.3816 | 0.3816 | 1.001 |
| ascad_desync_50 | 700 | 24064 | 0.014378 [0.012950, 0.015962] | 1.8800 | 0.4206 | 0.4206 | 0.986 |
| aes_hd | 1250 | 24576 | 0.005493 [0.004643, 0.006498] | 0.4919 | 0.0158 | 0.0159 | 0.8754 |
| aes_rd | 3500 | 11520 | 0.173958 [0.167145, 0.180989] | 5.4768 | 3.5024 | 3.5054 | 1.059 |
| dpav4 | 4000 | 1280 | 0.039844 [0.030433, 0.052008] | 3.3505 | 2.4078 | 2.4133 | 1.146 |
[... 9 lines omitted ...]
| ascad_desync_50 | 23.0741 | 5.06651e-10 | 0.00138606 | 0.00179074 |
| dpav4 | 20.7069 | 1.69928e-09 | 0 | 4.55626e-16 |

## Bounded conclusion

Datasets with a positive finite BI lower bound but an EPI interval touching zero: none.

Datasets where finite-benchmark BI is numerically above EPI and the hard-label accuracy's Wilson 95% lower e
    ndpoint is above random guessing: aes_hd, aes_rd, ascad_desync_0, ascad_desync_100, ascad_de …

Dimension by itself is not an EPI failure mode after model scores exist: EPI optimizes one scalar beta. BI i
    s better only in the narrower sense supported by the artifacts: it is calibration-free and i …

dataset,native_dimension,suite_size,selected_model,selection_records,calibration_records,evaluation_records,
    evaluation_successes,evaluation_accuracy,evaluation_accuracy_wilson95,accuracy_wilson_above_ …
ascad_desync_0,700,80,zaid_ascad_desync_0_no_preprocessing_6.hdf5,7680,7680,24064,129,0.005360704787234043,"
    (0.0045137672329648255, 0.006365540712814046)",True,0.45663840374561676,0.004733565821951712 …
ascad_desync_100,700,80,zaid_ascad_desync_100_no_preprocessing_5.hdf5,7680,7680,24064,359,0.0149185505319148
    93,"(0.013462476473456388, 0.016529471913595342)",True,1.933251182145414,0.3815748730661118, …
ascad_desync_50,700,80,zaid_ascad_desync_50_no_preprocessing_0.hdf5,7680,7680,24064,346,0.014378324468085107
```

```
    ,"(0.012949869722054237, 0.015961798987882074)",True,1.8800393759590872,0.4205990621860845,0 …
aes_hd,1250,80,zaid_aes_hd_no_preprocessing_2.hdf5,8192,8192,24576,135,0.0054931640625,"(0.00464321790550367
    8, 0.0064976781527344716)",True,0.4918530963296747,0.01575814402704539,0.8753504108831608,0. …
aes_rd,3500,80,zaid_aes_rd_no_preprocessing_9.hdf5,3584,3584,11520,2004,0.17395833333333333,"(0.167145098668
    29977, 0.18098893920063583)",True,5.476813696865533,3.5024283358737023,1.0588027182551607,3. …
dpav4,4000,80,zaid_dpav4_no_preprocessing_9.hdf5,256,256,1280,51,0.03984375,"(0.03043337380474899, 0.0520078
    4827411856)",True,3.350497247084133,2.407818321020527,1.1459884593011762,2.4133203749520815, …
[exit code 0, 1 s]
```

What to look for

In `temperature_stress.csv`, the `raw_pi` column swings widely across the rows (by 4 bits on ASCAD, by 27 bits on AES-RD), the `epi` column stays the same to eight decimals, and the last column says that every guess stayed the same. That is the point: BI and EPI do not care how confident the model is, only what it guesses. Raw PI does.